



September 2020 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates - SOLUTIONS

Instructions to candidates

This paper contains 7 questions. Your answers to **all 7 questions** will count towards the final mark. The number of marks for each question is indicated next to the question.

Answers should be justified by showing work.

Time Allowed Submit PDF with answers within 48 hours after official start of exam
Expected effort required: 2 hours

You are supplied with: *No additional material*

You may also use: *Open book exam*

Calculators: *Calculators are allowed in this examination (open book)*

1. **(8 points – 10 minutes)** In a tennis match with a score at deuce, the game is won by the first player who gets a clear lead of two points (i.e. wins the next two points). If the probability that given player wins a particular point is θ , and all points are played independently, what is the probability that this player eventually wins the game?

Let p_1 (p_2) indicate whether a given player wins the first (second) point.

$$\Pr(p_1 = \text{yes}, p_2 = \text{yes}) = \theta^2$$

$$\Pr(p_1 = \text{yes}, p_2 = \text{no} \cup p_1 = \text{no}, p_2 = \text{yes}) = 2\theta(1 - \theta) \text{ (clearly smaller than 1 - needed for cgce geometric sequence below)}$$

$$\Pr(\text{two consecutive points won}) = \theta^2 + 2\theta(1 - \theta)(\theta^2 + 2\theta(1 - \theta) + (\theta^2 + 2\theta(1 - \theta))^2 + \dots) \text{ (geometric sequence)}$$

$$\text{So, } \Pr(\text{two consecutive points won}) = \theta^2 \left(\frac{1}{1 - 2\theta(1 - \theta)} \right)$$

(Answer similar to football example in quiz 1)

Points should be awarded for partial but relevant statements. (e.g, 2 **points** for θ^2 (prob two consecutive points), 2 **points** for $2\theta(1 - \theta)$ (prob one point won and one lost - back to deuce).

2. **(8 points – 10 minutes)** Let the discrete random variable X have probability function

$$f_X(x) = \begin{cases} p & x = -1, 1 \\ 1 - 2p & x = 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $0 < p < 1/2$. Derive the coefficient of skewness and excess kurtosis of X as a function of p . Interpret your results.

2 points: for definition of skewness and kurtosis

$$\begin{aligned} \text{coefficient of skewness} &: \frac{E[(X - \mu)^3]}{\sigma^3} \text{ and} \\ \text{coefficient of excess kurtosis} &: \frac{E[(X - \mu)^4]}{\sigma^4} - 3 \end{aligned}$$

4 points: All required computations

$$\begin{aligned} E(X) &= -1 \cdot p + 1 \cdot p + 0 \cdot (1 - 2p) = 0 \\ E(X^2) &= (-1)^2 \cdot p + (1)^2 \cdot p + (0)^2 \cdot (1 - 2p) = 2p \equiv \sigma^2 \\ E(X^3) &= (-1)^3 \cdot p + (1)^3 \cdot p + (0)^3 \cdot (1 - 2p) = 0 \\ E(X^4) &= (-1)^4 \cdot p + (1)^4 \cdot p + (0)^4 \cdot (1 - 2p) = 2p \end{aligned}$$

hence

skewness: = 0 (symmetric)

$$\text{kurtosis:} = \frac{2p}{(2p)^2} - 3 = \frac{1}{2p} - 3$$

2 points: interpretation. The distribution is symmetric, and when $p < 1/6$ will we have fatter tails than that of a standard normal.

3. **(8 points – 10 minutes)** Let X have the probability density function given by

$$f_X(x) = \theta x^{\theta-1}, 0 < x < 1 \\ = 0 \text{ elsewhere,}$$

with $\theta > 0$. Derive the probability density function of the random variable Y where $Y = X^2 - 1$.

2 points: First of all we note that over its domain it is a one-to-one transformation.

With $u(x) = x^2 - 1$, the inverse is given by $v(y) = (y + 1)^{1/2}$

5 points: $g_y(y) = f_x(v(y)) \cdot |v'(y)|$ for $-1 < y < 0$. With $|v'(y)| = \frac{1}{2}(y + 1)^{-1/2}$, we obtain

$$g_y(y) = \theta \left((y + 1)^{1/2} \right)^{\theta-1} \times \frac{1}{2} (y + 1)^{-1/2} = \frac{1}{2} \theta (y + 1)^{\theta/2-1} \text{ for } -1 < y < 0 \\ = 0 \text{ elsewhere}$$

(students loose 3 points for omitting relevant ranges)

4. **(10 points – 12 minutes)** Let X and Y be independent random variables, with marginal density functions:

$$f_X(x) = \frac{x}{2}, 0 \leq x \leq 2, 0 \text{ elsewhere}$$

$$f_Y(y) = 1, 0 \leq y \leq 1, 0 \text{ elsewhere}$$

Find the joint cumulative distribution function for all possible values of x and y .

Here $F(x) = \Pr(X \leq x) = x^2/4$, $0 \leq x \leq 2$, $F(y) = \Pr(Y \leq y) = y$, $0 \leq y \leq 1$, and by independence $F(x, y) = x^2 y/4$. So we have

	$y < 0$	$y \in (0, 1)$	$y > 1$
$x < 0$	0	0	0
$x \in (0, 2)$	0	$\frac{x^2 y}{4}$	$\frac{x^2}{4}$
$x > 2$	0	y	1

Those that don't state that $F(x, y) = 0$ for $x < 0$ or $y < 0$ lose **2 points**.

Those that don't state that $F(x, y) = 1$ for $x > 2$ AND $y > 1$ lose **1 point**.

Those that don't state that $F(x, y) = y$ for $x > 2$ and $y \in (0, 1)$ (or $y > 1$ and $x \in (0, 2)$) lose **2 points**.

Thus, for those simply report $F(x, y) = x^2y/4$ obtain **6 points**.

5. **(6 points – 7 minutes)** Let $Y = aX + b$, where a and b are fixed constants and $X \sim N(\mu, \sigma^2)$. Derive $E(Y)$, $E(Y|X = x)$, $Var(Y)$ and $Var(Y|X = x)$.

- **1 point.** By properties of the expectation operator $E(Y) = aE(X) + b = a\mu + b$
- **1 point.** By properties of the conditional expectation operator $E(Y|X = x) = ax + b$
- **2 point.** $Var(Y) = E((Y - E(Y))^2) = E(a^2(X - E(X))^2) = a^2Var(X) = a^2\sigma^2$
- **2 point.** By definition of conditional variance $Var(Y|X = x) = E(Y^2|X = x) - E(Y|X = x)^2 = E((aX + b)^2|X = x) - (ax + b)^2 = (ax + b)^2 - (ax + b)^2 = 0$

6. **(30 points - 36 minutes)** Let X is a random variable from the Pareto distribution, which has density

$$f(x) = \frac{\theta}{x^{\theta+1}}, \quad x > 1, \quad 1 < \theta < \infty$$

Let $X_1, \dots, X_n$ be a random sample from this distribution.

- (a) 12 points Derive the maximum likelihood estimator for θ ($\hat{\theta}_{MLE}$).

By independence the likelihood can be expressed as

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n f(x_i) \\ &= \frac{\theta^n}{x_1^{\theta+1} \dots x_n^{\theta+1}} \\ \log L(\theta) &= n \log(\theta) - (\theta + 1) \sum_{i=1}^n \log(x_i) \\ \text{F.O.C.: } \frac{\partial \ln L}{\partial \theta} &: \frac{n}{\hat{\theta}_{MLE}} - \sum_{i=1}^n \log(x_i) = 0 \\ \text{or } \hat{\theta}_{MLE} &= \left(\frac{1}{n} \sum_{i=1}^n \log(x_i) \right)^{-1} \quad \text{estimate for a given sample} \end{aligned}$$

The MLE estimator therefore equals

$$\hat{\theta}_{MLE} = \left(\frac{1}{n} \sum_{i=1}^n \log(X_i) \right)^{-1}$$

6 points: Provide the likelihood function (subtract 2 points not referred to use of independence)

6 points: FOC (properly using $\hat{\theta}_{MLE}$) and solve (subtract 2 points if θ is used instead because FOC only need to be satisfied for $\hat{\theta}_{MLE}$)

- (b) (8 points) Show that the expected value of $\log(X)$ equals θ^{-1} .

Hint: You will have to make use of integration by parts and may want to make use of the result that, here, $\lim_{n \rightarrow \infty} \frac{\log x}{x^\theta} = 0$. Partial credit will be awarded for correct approach.

By definition

$$\begin{aligned} E(\log(X)) &= \int_1^\infty \underbrace{\log(x)}_u \underbrace{\theta x^{-(\theta+1)}}_{v'} dx \\ &= \left[\underbrace{\log(x)}_u \cdot \underbrace{-x^{-\theta}}_v \right]_1^\infty - \int_1^\infty \underbrace{x^{-1}}_{u'} \cdot \underbrace{-x^{-\theta}}_v dx \\ &= \int_1^\infty x^{-(\theta+1)} dx \text{ using result given in hint} \\ &= \left[-\frac{1}{\theta} x^{-\theta} \right]_1^\infty \\ &= \theta^{-1} \text{ as } \theta > 1 \end{aligned}$$

- (c) (10 points) Show that the ML estimator for θ is consistent. Properly annotate your answer.

4 points Answers with clear reference to Slutsky and Law of Large numbers

6 points Using Slutsky we note $\text{plim } \hat{\theta} = (\text{plim } \frac{1}{n} \sum_{i=1}^n \log(X_i))^{-1}$ provided $\text{plim } \frac{1}{n} \sum_{i=1}^n \log(X_i) \neq 0$. By LLN (Kinchine) we know $\text{plim } \frac{1}{n} \sum_{i=1}^n \log(X_i) = E(\log(X_i))$ (sample average converges to its population analogue). Therefore

$$\text{plim}(\hat{\theta}_{ML}) = \frac{1}{\text{plim } \frac{1}{n} \sum_{i=1}^n \log(X_i)} = \frac{1}{\theta^{-1}} = \theta$$

7. Let $X_1, \dots, X_{100}$ denote a random sample of 100 observations from $N(\theta, 4)$. You want to test the hypothesis that $H_0: \theta = 5$ against the alternative $H_A: \theta > 5$. For a given sample, you obtain a point estimate $\bar{x} = 5.2$.

- (a) Discuss how you would test this hypothesis using a 5% level of significance.

- Use the Z-test: $Z = \frac{\bar{X} - 5}{\text{Stdev}(\bar{X})} \sim N(0, 1)$ under H_0 , where $\text{Stdev}(\bar{X}) = \sqrt{\frac{\sigma^2}{n}} = \sqrt{\frac{4}{100}} = \frac{2}{10}$.

- We should reject if realisation of Z statistic exceeds 1.645 (5% level of significance $N(0,1)$ critical value)
 - Here our realisation of our test statistic equals $\frac{5.2-5}{2/10} = 1.0$. Since it does not exceed 1.645 we will not reject the null hypothesis at the 5% level of significance
- (b) *What is associated p-value of our one-sided test, and what conclusion can you draw from it?*
- To obtain its p-value we compute $\Pr(Z > 1.0) = .1587$
 - This tells use that we would be able to start rejecting the null when the significance level equals 15.87% (at the 5% we cannot reject the null).
- (c) *Discuss how you can obtain the power of this test at θ_1 where $\theta_1 > 5$. Interpret your finding.*
- Need to evaluate $\Pr\left(\frac{\bar{X}-5}{\text{Stdev}(\bar{X})} > 1.645 \mid \theta = \theta_1\right) = \Pr\left(\frac{\bar{X}-\theta_1}{\text{Stdev}(\bar{X})} > 1.645 + \frac{5-\theta_1}{\text{Stdev}(\bar{X})} \mid \theta = \theta_1\right)$
 Since $\frac{\bar{X}-\theta_1}{\text{Stdev}(\bar{X})} \sim N(0, 1)$ under the alternative, this equals $1 - \Phi\left(1.645 + \frac{5-\theta_1}{\text{Stdev}(\bar{X})}\right) = 1 - \Phi\left(1.645 + \frac{10(5-\theta_1)}{2}\right) = \Phi(5\theta_1 - 26.645)$.
 - Interpretation: The further away θ_1 is to the hypothesized value, the closer this probability (power, our ability to reject the null when it is false) will be to one.
- (d) *It is argued that performing this test is more powerful than conducting a two-sided test at the same level of significance. Explain this comment intuitively and briefly indicate how you can obtain the power at θ_1*
- Intuitively: Performing a one-sided test at the same level of significance will require us to start rejecting earlier. As that also means that we start rejecting the null when it is false, we will have more power using a one-sided than a two-sided alternative. Power relates to our ability to reject the null when it is false.
 - In this case we need to evaluate

$$\Pr\left(\left(\frac{\bar{X}-5}{\text{Stdev}(\bar{X})} > 1.96\right) \cup \left(\frac{\bar{X}-5}{\text{Stdev}(\bar{X})} < -1.96\right) \mid \theta = \theta_1\right) =$$

$$\Pr\left(\frac{\bar{X}-5}{\text{Stdev}(\bar{X})} > 1.96 \mid \theta = \theta_1\right) + \Pr\left(\frac{\bar{X}-5}{\text{Stdev}(\bar{X})} < -1.96 \mid \theta = \theta_1\right).$$
 Using the same ideas as in (c), this equals

$\Pr\left(\frac{\bar{X}-\theta_1}{\text{Stdev}(\bar{X})} > 1.96 + \frac{5-\theta_1}{\text{Stdev}(\bar{X})} \mid \theta = \theta_1\right) + \Pr\left(\frac{\bar{X}-\theta_1}{\text{Stdev}(\bar{X})} < -1.96 + \frac{5-\theta_1}{\text{Stdev}(\bar{X})} \mid \theta = \theta_1\right)$. for
 which we can again use the fact that with $\theta = \theta_1$, $\frac{\bar{X}-\theta_1}{\text{Stdev}(\bar{X})}$ has a $N(0,1)$ distribution.